

Experiment 03 — Cookies, Sessions and Authentication

GitHub-readable preview of the presentation. Download the matching .pptx for the editable PowerPoint.

1. Syllabus tasks

- Display a login form.
- Check demo username and password.
- Store the logged-in user in a session.
- Store a role value in a cookie.
- Protect the profile route.
- Destroy the session during logout.

2. Workflow

Experiment 03 — Cookies, Sessions and Authentication



Workflow: input → processing → data/state → output

3.1. Code pathway — app.js

```
1 /**
2  * Experiment 03: ExpressJS Cookies, Sessions and Authentication
3  *
4  * Demo credentials:
5  * Username: student
6  * Password: 1234
7  *
8  * What it demonstrates:
9  * - Form-data handling
10 * - Cookies
```

3.2. Code pathway — app.js

```
11 * - Sessions
12 * - Login and logout
13 * - Protected profile route
14 */
15
16 const express = require('express');
17 const cookieParser = require('cookie-parser');
18 const session = require('express-session');
19
20 const app = express();
```

4. Runtime walkthrough

Runtime walkthrough — Experiment 03 — Cookies, Sessions and Authentication



Runtime: start → enter → execute → observe

5. Run and observe

- Start the server / Vite app or open the MongoDB shell as described in the experiment README.
- Enter the sample input or call the required route.
- Watch the terminal for logs and errors.
- Verify the browser, API response, or database result.

6. Viva questions

- What is the purpose of this experiment?
- Which input is required?
- Which line or command performs the main operation?
- What output should you observe?
- What happens when the input changes?